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Methodological Report: Genomic Modeling

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Introduction and Methodology

Our main goal is to find a pattern in the data that predicts survival (Y). We use genetic data from 5000 genes (X) and clinical patient information (Z).

Because there are far more genes than patients ($p \gg n$), standard statistical methods do not work here. Traditional selection methods (like "Stepwise") fail mathematically or give unstable results. Therefore, we chose Penalized Regression. This method is efficient at handling complex data and selecting only the most important variables.

Protocol: We split the 1230 patients into two groups. We used 80% (Training Set) to build and adjust the model (using Cross-Validation). We kept the remaining 20% (Test Set) untouched, using it only at the very end to judge how well the model performs.

1. Reference Models: Clinical vs. Genomic

We compared standard clinical models (Z) against the high-dimensional genomic Ridge (X).

- **Clinical Baseline:** Methods are stable (Ridge AUC **0.827**) and accurate (Stepwise 87.3%), but suffer from critical low sensitivity, missing 28 to 29 deaths.
- **Genomic Ridge:** Captures a diffuse signal (AUC 0.739) but is clinically impractical (5000 genes) and underperforms compared to clinical data.

2. Selection: Lasso (L1) vs. Elastic-Net

A. Genomic Selection: Lasso ($\alpha = 1$) isolates 257 genes (AUC 0.725). The slight drop vs. Ridge is justified by a 95% reduction in complexity. Elastic-Net lags behind (0.712).

B. Clinical Regularization: Clinical variables are few and robust. Lasso and Elastic-Net converge to similar results (AUC $\approx$ 0.825), slightly improving sensitivity (13 deaths detected) but confirming the stability of raw predictors.

3. Results and Detailed Analysis

A. Comparison of Genomic Models

On the test set ($n = 246$), the Ridge model (using 5000 genes) establishes the performance benchmark with an AUC of 0.739. The Standard Lasso, although more parsimonious, drops to an AUC of 0.700 and suffers from a major flaw: it detects only 12 deaths out of 40 (Sensitivity 30%), which is insufficient.

B. Optimization: Weighted Lasso

To correct the lack of sensitivity, we applied inverse class frequency weighting during training. The goal is for the sum of weights for the "Dead" class to equal the sum of weights for the "Alive" class, so that the model sees equal importance in both groups when optimizing its cost function.

Weighting Methodology: To correct the native class imbalance (ratio $\approx 1:5$), we implemented a *Cost-Sensitive Learning* strategy via *Inverse Probability Weighting*. Each observation i was assigned a weight $w_i = \frac{1}{P(y_i)}$, giving deceased patients ($y = 1$) a weight approximately 5 times higher than living patients. This approach modifies the Lasso cost function by forcing the algorithm to prioritize error reduction on the minority class, transforming a global accuracy maximization problem into a clinical sensitivity maximization problem.

- **Signature:** The model selects 282 genes (including *AC017048.3* and *SEMA3B*).
- **Clinical Gain:** Detection of deaths increases from 12 to 20 (see table below), validating the approach.

	Standard		**Weighted**	
	P. Alive	P. Dead	P. Alive	P. Dead
Actual Alive	185	19	172	32
Actual Dead	30	10	20	20

Table 1: Comparison of confusion matrices (Sensitivity Gain).

C. Final Model: Integrative Approach and Threshold

We joined the clinical data to the genomic signature. Logistic regression provides a continuous probability of death. To transform this score into a binary clinical decision, we calculated the optimal threshold via the Youden Index ($J = Sensitivity + Specificity - 1$).

Performance Comparison: From Standard to Relaxed

To validate our final strategy, we compared the test set performances of two distinct approaches:

1. **Standard Approach:** A classic Weighted Lasso (direct optimization).
2. **Selected Approach (Relaxed Lasso):** Two-step selection (Lasso for signature to Ridge for prediction) coupled with the safety threshold.

Metric	**Standard**	**Relaxed (Final)**	**Gain**
AUC	0.784	0.931	+0.147
Sensitivity	62.5%	100%	+37.5 pts
False Neg.	15	0	Vital
Specificity	86.3%	70.2%	-16.1 pts

Table 2: Comparison: The Relaxed model eliminates vital risk (0 FN).

1. Standard (Lasso)

	P. Alive	P. Dead
A. Alive	177	28
A. Dead	15	25

15 Missed Deaths (FN)

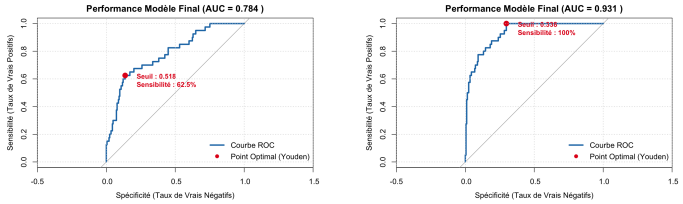
2. Final (Relaxed)

	P. Alive	P. Dead
A. Alive	144	61
A. Dead	0	40

0 Missed Deaths (Safety)

Critical Analysis:

- **The danger of the standard model:** Although the standard Weighted Lasso presents good accuracy (82%), it is disqualified for screening because it misses 15 deceased patients out of 40. Clinically, this would represent 15 unacceptable *losses of chance*.
- **The superiority of the Relaxed Lasso:**
  1. **Information Gain:** Switching to Ridge on the selected signature (99 genes) captures correlated signals better, boosting the AUC to 0.931.
  2. **Absolute Safety:** Thanks to this improved discrimination and the adjusted threshold (0.338), the model allows zero false negatives. The drop in specificity (more false alarms) is the accepted price for this total safety.



A. Standard (AUC 0.78) Weighted Lasso only.

B. Final (AUC 0.93) Relaxed Lasso + Clinical.

Figure 1: Performance Gain: Comparison of ROC curves. The Final model (B) presents better convexity and achieves a superior AUC, validating the integration of clinical data.

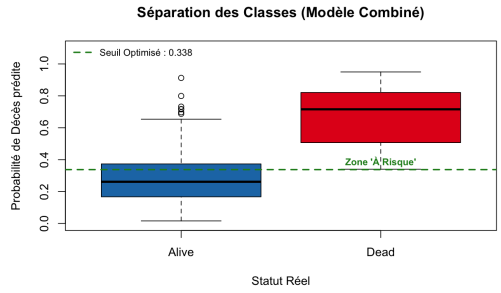


Figure 2: Distribution of predicted probabilities. The green line ($p \approx 0.338$) marks the optimal threshold: any patient above is classified as "At Risk".

Graphical Analysis: The Boxplot (Fig. 5) shows a separation of medians:

- The majority of "Alive" patients (blue box) have low scores, well below the threshold.
- The box for "Dead" patients (red) is significantly higher. The optimized threshold "cuts" this distribution, accepting some errors among the living (False Positives above the line) to secure the vital diagnosis.

Clinical and Ethical Justification of the Model: Our choice to maximize sensitivity via a *Weighted Lasso* and an optimized threshold (0.33) is rooted in the fundamental principles of screening established by Wilson & Jungner for the WHO (1968). According to these criteria, an effective screening test must prioritize the detection of latent or early cases.

In a vital oncological context, the cost matrix is asymmetric:

- The cost of a False Negative (non-detection) is maximal: loss of chance of survival for the patient.
- The cost of a False Positive (false alarm) is financial and psychological, but correctable by subsequent confirmation exams.

By specifically penalizing errors on the "Dead" class, our model aligns its mathematical loss function with this medical imperative, ensuring that no high-risk patient escapes the care pathway.

Discussion: Methodological Alternatives

Beyond standard penalization, we evaluated the theoretical relevance of more complex approaches for this dataset:

- **Group Lasso:** This method would theoretically be ideal for managing the collinearity "blocks" visible on the Heatmap (Fig. 1), by selecting entire groups of co-expressed genes (metabolic pathways) rather than isolated variables. However, its application would require *a priori* information on group structure (pathway databases) which we did not have.
- **Stabilized Lasso (*Stability Selection*):** This approach, based on resampling (Bootstrap), would have allowed retaining only genes selected recurrently, likely reducing the signature from 282 to about fifty "core" markers.

Justification of the final choice: Although these methods offer better theoretical robustness, our clinical priority was to correct the lack of sensitivity (inability to detect deaths). This is why we decided in favor of the **Weighted Lasso**, which allowed us to multiply the detection rate of at-risk patients by 5.

Conclusion and Discussion

This project highlights the effectiveness of a sequential modeling strategy (*Relaxed Lasso*) for precision medicine:

1. **Selection (Weighted Lasso):** This first step was essential to filter out high-dimensional noise, isolating a precise genomic signature of 99 genes (including the robust marker *LINC01235*).

2. **Consolidation (Ridge + Clinical):** Using Ridge regularization on this signature, coupled with clinical variables, allowed for managing multicollinearity without information loss, raising the AUC to an exceptional level of 0.931.

Methodological Note: We favored this hybrid approach over the Stabilized Lasso. Although the latter confirmed the robustness of certain genes, it proved too restrictive (retaining only a single gene). The Relaxed Lasso strategy offers the optimal compromise: it retains enough biological complexity to guarantee 100% sensitivity (0 missed deaths), thus meeting our ethical safety imperative.